



UPSC Prelims

Science & Tech

Class No. 9

Space Technology

Indian Launch Vehicle

1. Historical

- SLV : Satellite launch vehicle
- ASLV : Augmented Satellite launch vehicle

2. Operational

- Sounding rockets : Every Wednesday tested in thumba (60s) :
Sounding rocket means Taking measurement
- PSLV (1993)
- GSLV (2001)
 - Mark - 3 developed recently

PSLV and GSLV

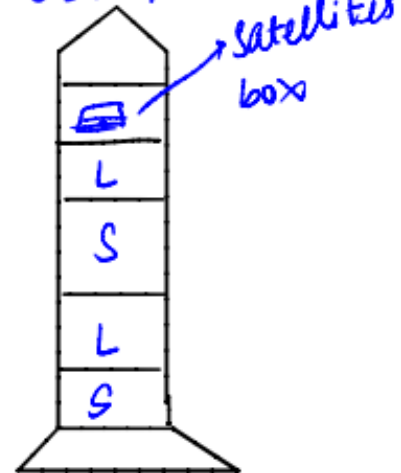
PSLV	GSLV
• Polar satellite launch vehicle • Can be used for lighter satellites only • Remote sensing satellites in polar sun synchronous orbit	• Geosynchronous satellite launch vehicle . • Can be used for heavier satellites also (2 ton-4 ton) • Heavier Communication Satellite in Geosynchronous transfer orbit

PSLV

4 Stage:

Solid → Liquid → S → L

HTPB — UDMH



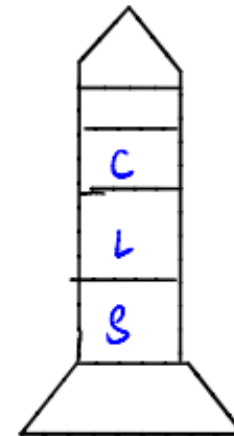
3 versions:

(1) Core alone
(without strap on
motors)

GSLV

3 stage

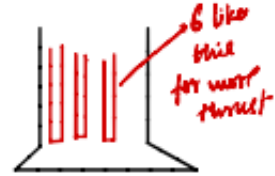
Solid → Liquid →
cryogenic



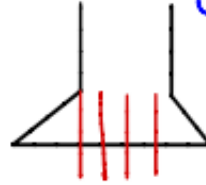
versions

(1) Mk-I
(2 tons)

(i) standard config
↓
with strap on motors
(6)

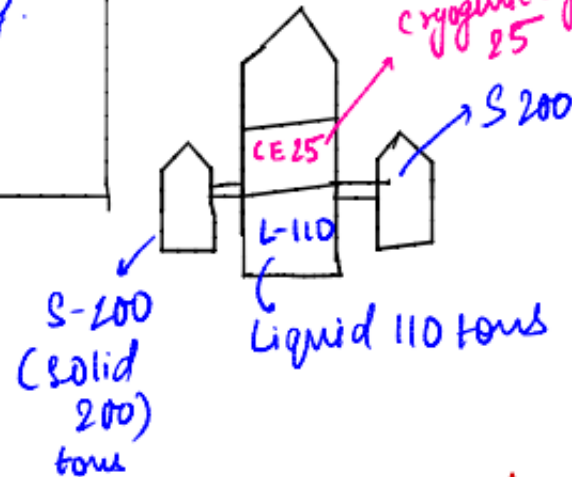


(ii) Mark - II
4 strap on motors
(3 ton)



(iii) extra large
(6) strap on motors but
more fuel capacity.

(iii) Mark - III



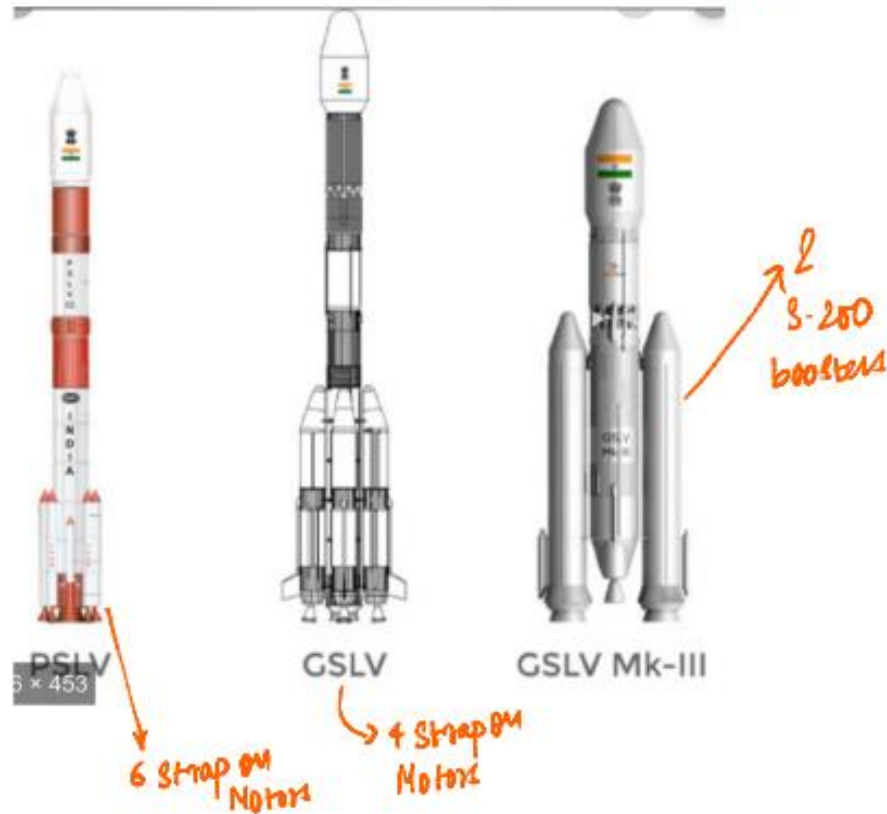
→ these are heavy boosters.
using solid fuel.

process of MK-3

S-200 → L-110 → CE-25

S-200

capacity of MK-III : 4 ton to 5 ton.



GSLV MK-III

- Heaviest launch vehicle of India (640 tones) (42 inter long)
- Can launch 4000-5000 kg class satellites
- 3 stages launch
 - S-200 - boosters (Solid)
 - L-110 - Liquid
 - CE - 25 Cryogenic

Importance

- Heavy communication Satellites (GSAT)
- Saves India's foreign exchange
 - Makes us self reliant
- Help each foreign exchange by launching foreign satellites
 - (Int space **sunset**) India **arriu** they each only 1/. Of world space revue .

PSLV in Current Affairs

- PSLV on its 55th mission successfully launched 3 Singaporean satellites in the second commercial mission of NSIL.
- Besides this, ISRO also achieved successful launch of PSLV orbital experimental module (POEM)

PSLV in Current Affairs

About POEM: A platform which allows in-orbit scientific experiments using the final (fourth) stage of PSLV.

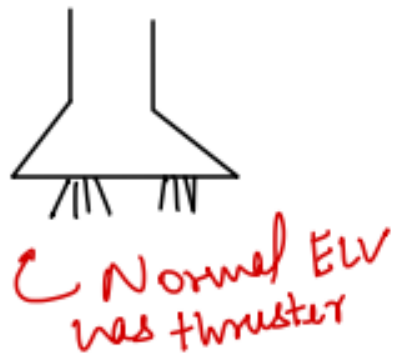
- Dedicated Navigation Guidance and Control system: Act as platform's brain to stabilize it with specified accuracy.
- Power source: Mounted solar panels and a Li-Ion battery.

ISRO's Launch Vehicles

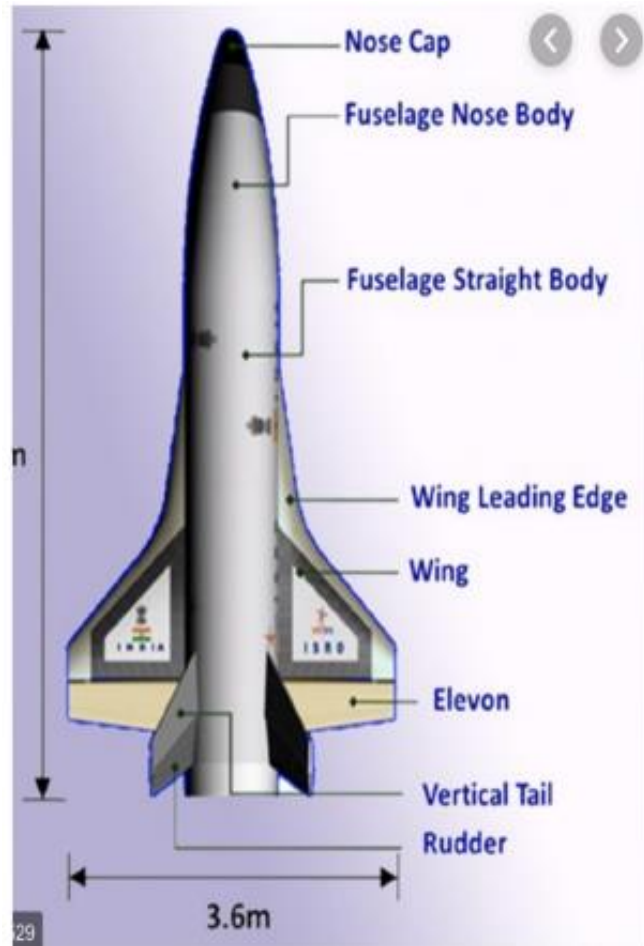
Launch Vehicle	Stages	Payload capacity and orbits
Small Satellite launch Vehicle (SSLV)	3 stages Launch Vehicle (Three Solid Propulsion stages and liquid propulsion based velocity module as a terminal stage)	Mini, Micro, or Nanosatellites (10 to 500 kg mass) to a 500 km orbit.
Polar Satellite Launch Vehicle (PSLV)	Four stage launch vehicle (1st & 3rd stage; Solid, 2nd & 4th stage; liquid)	Up to 1750 kg to Sun synchronous Polar orbits, 1425 into Geosynchronous and Geostationary orbits.
Geosynchronous Satellite Launch Vehicle (GSLV) Mark II	Three stage launch vehicle (1st solid, 2nd liquid, 3rd Cryogenic upper stage)	Up to 2250 into geosynchronous transfer orbits up to 6 tonne in low earth orbits
Geosynchronous Satellite Launch Vehicle MK-III (LVM3)	Three stage launch vehicle (1st liquid, 2nd solid, 3rd Cryogenic, Upper stage)	Up to 4 tonne into Geosynchronous Transfer orbits, 8 tonne into low earth orbits.

RLV : Reusable Launch Vehicle

- RLV uses a winged air-craft which re enters and returned back to the earth
- Use Boosters to go to outer space.
- 2 stage to orbit launch vehicle



ISRO Proposed RLV



Advantage of RLV for India:

It is like Nasa's space shuttle. NASA not using space shuttle and using private companies - Falcon - 9 by Space-X

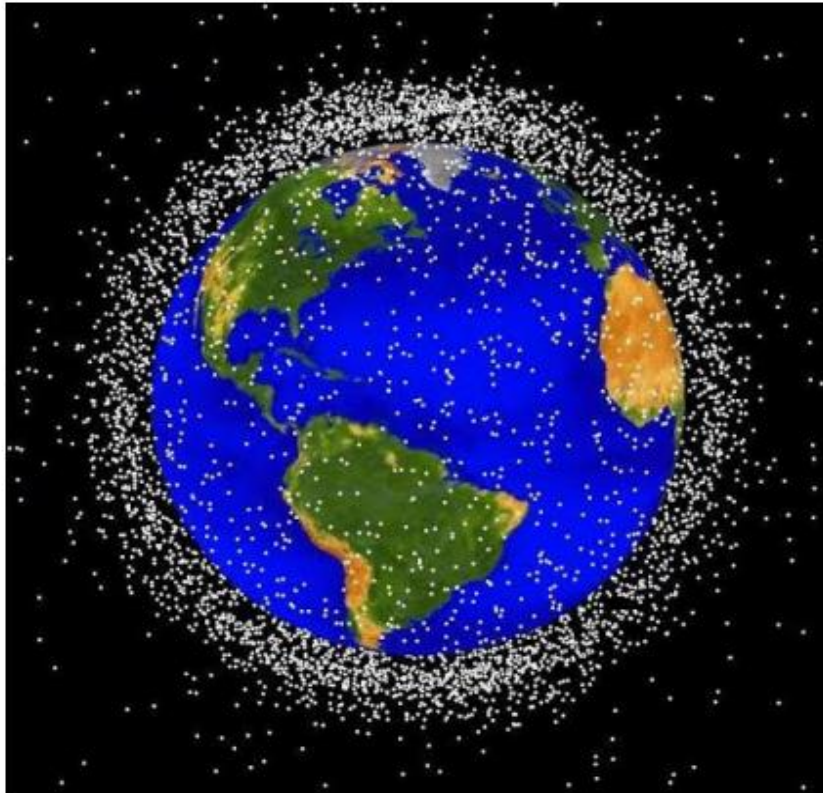
- Commercial launch of satellites will be low cost. (RLV is more cost effective wrt ELV)
- International space market - Capture foreign reserve.
 - Compete with space X
- Space tourism - India has cheaper technology than other nations

Challenges for RLV Development

For Re-Entry of RLV : 10km runway is needed.

- **Infrastructure** challenges: Long runway needed on coast to avoid inland disasters. India's undulating coastline will be an issue.
- **Cost benefit** of investing in RLV instead of Socio-Economic Programs will be questioned. As it is extremely expensive .
- **Logistic** Challenges : Higher Logistic costs
- **Training** Challenges : Training and maintenance cost.
- By the time India is able to develop its RLV, Space X might already capture the market. Competition from International space entities. (Falcon-9 Heavy)

Space Debris



Space Debris

- Why in News? : China rocket debris fell in Indian Ocean near Maldives.
- Parts of Long March 5B re-entered the atmosphere and landed in the Indian Ocean, with the bulk of its components destroyed upon re-entry into the Earth's atmosphere. Long March 5B was carrying components for the construction of China's space station, which will be completed by the end of next year.
- Space debris encompasses both natural (meteoroid) and artificial (man-made) particles. Meteoroids are in orbit about the sun, while most artificial debris is in orbit about the Earth which is commonly referred to as orbital debris.

Space Debris

- Much of the debris is in low Earth orbit (LEO), though some debris can be found in geostationary orbit.
- There are about 500,000 pieces of debris the size of a marble or larger orbiting the Earth, travelling at speeds up to 17,500 mph.
- International guidelines suggest removing space crafts from LEO within 25 years of the end of their mission. However, only 60 percent of missions follow the guidelines.

Tackling Space Debris

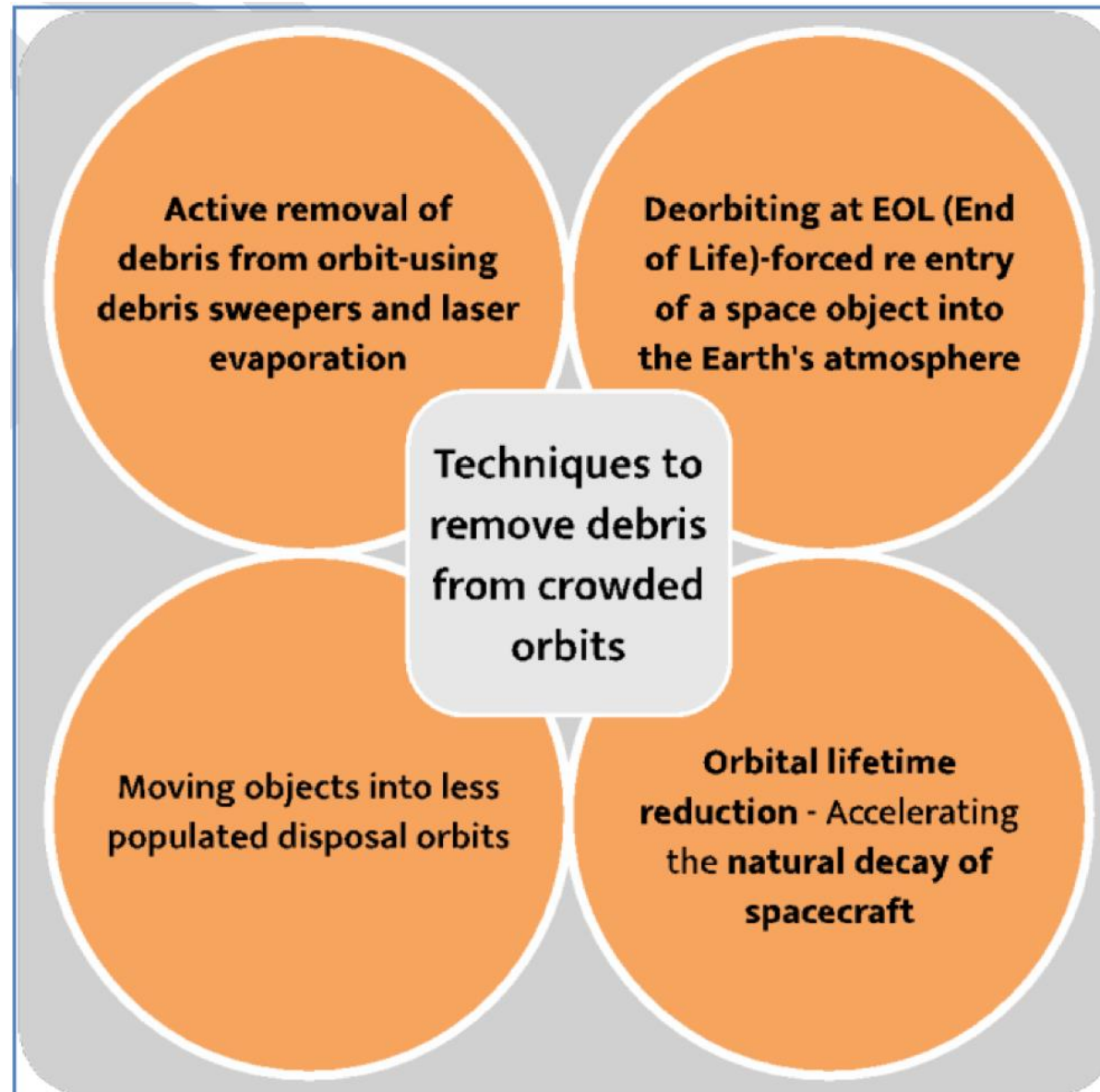
- Mitigating Damage: Space debris is tracked by a number of countries, including Germany, France, UK and USA.
- ISRO has come up with ‘Project NETRA’ - an early warning system in space to detect debris and other hazards to Indian satellites.
- Indraprastha Institute of Information Technology Delhi is working on a project titled “Orbit computation of Resident Space Objects for Space Situational Awareness” to predict collision from space debris.

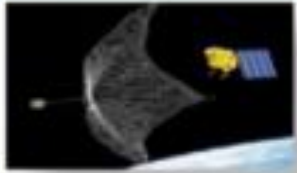
Tackling Space Debris

- Avoiding future debris: by adopting better designs of rockets and other objects. For example, making rockets reusable could vastly cut down waste. o UK's TechDemoSat-1 (TDS-1), was designed in such a way that once its mission is over, a system, would drag the satellite to re-enter the atmosphere and burn up.

Tackling Space Debris

- Removal of the debris:
- End-of-Life Services by Astroscale Demonstration (Elsa-D) was launched to locate and retrieve used satellites and other space junk.
- Remove Debris is an EU research project to give in orbit demonstrations of cost-effective technologies that can be used to observe, capture and dispose of space debris. It has performed key technology demonstrations including: Net Capture, Harpoon Capture,





Net Capture

It involves a net that will be deployed at the target CubeSat.



Harpoon Capture

Which will be launched at a target plate made of "representative satellite panel materials".



De-orbiting process

As it enters Earth's atmosphere, the spacecraft will burn up, leaving no debris behind.



Vision-Based Navigation

Using cameras and LiDAR, the platform will send data about the debris back to the ground for processing

Way forward for space junk

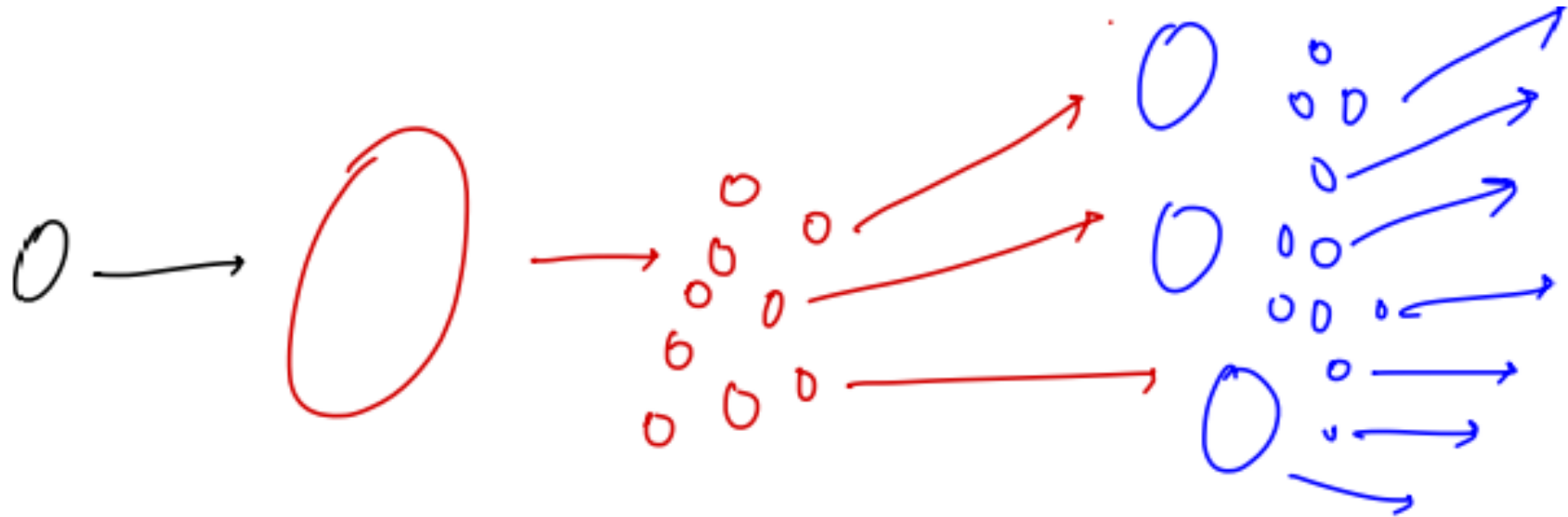
- 1. Kumotori Experiment (White Stork Jexa):** Use of tether to attach space junk magnetic tether) once tethered, it will be pulled down to atmosphere and burnt out by air friction. Japan tried it in 2017, but experiment FAILED.
- 2. Remove debris project :** Harpoon Satellite to catch debris (net capture - of space debris)
3. Providing high thrust and escape velocity to send to outer universe.
4. Production of Satellite that do not produce debris -
Ex. RLV even satellites need to be reusable.

Kessler Syndrome: Donald Kessler of NASA.

- Collision in space generates fragments that will collide again, leading to a cascade of collisions and proliferation of debris
- Outer space treaty by UN trying to clear space.

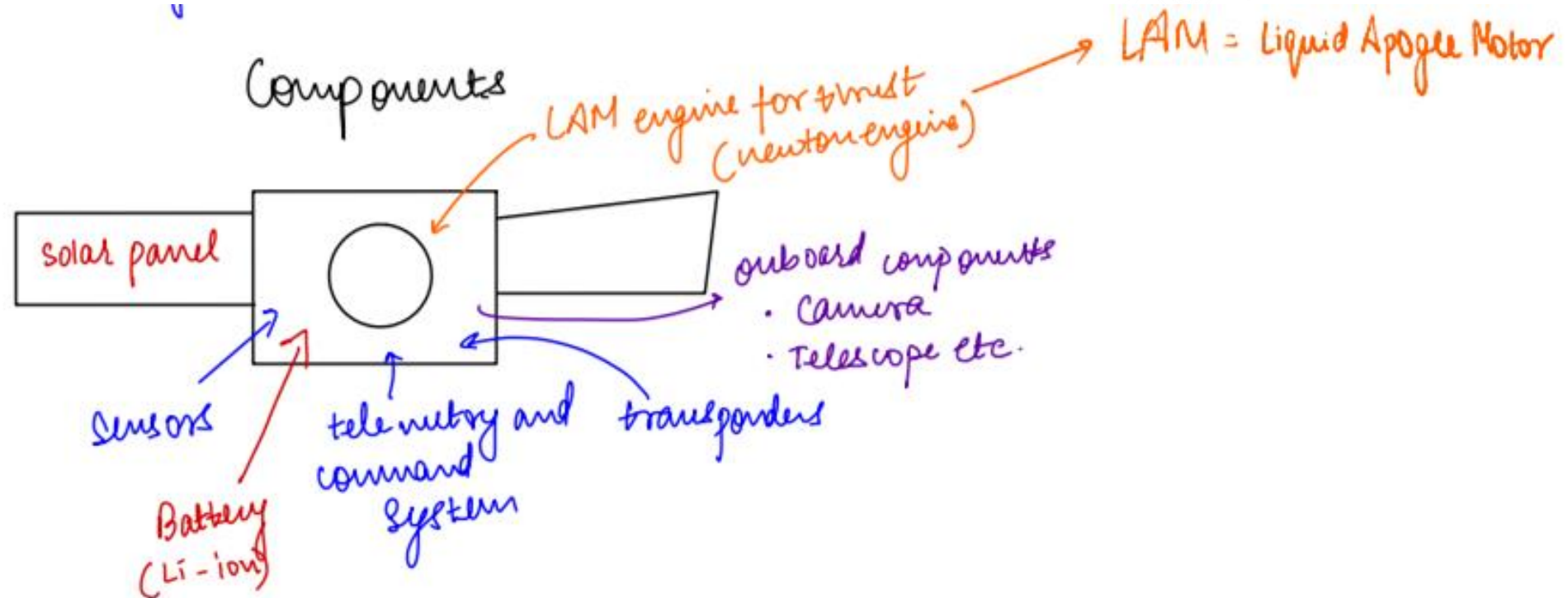
UNOSA : UN office on outer space activities was set up committee to investigate space debris.

- Space debris with large speed are the most dangerous as there is maximum chance of collisions.



SATELLITES

- Space crafts designed to carry payloads such as transponders , covers, instruments for various space applications.
- Different types of satellites for applications :
 - Communication
 - Meterology
 - Astronomy
 - Remote Sensing
 - Navigation
 - Defence and Reconnaissance
 - Anti Satellites



Transponders:

Transmitters and receiver.

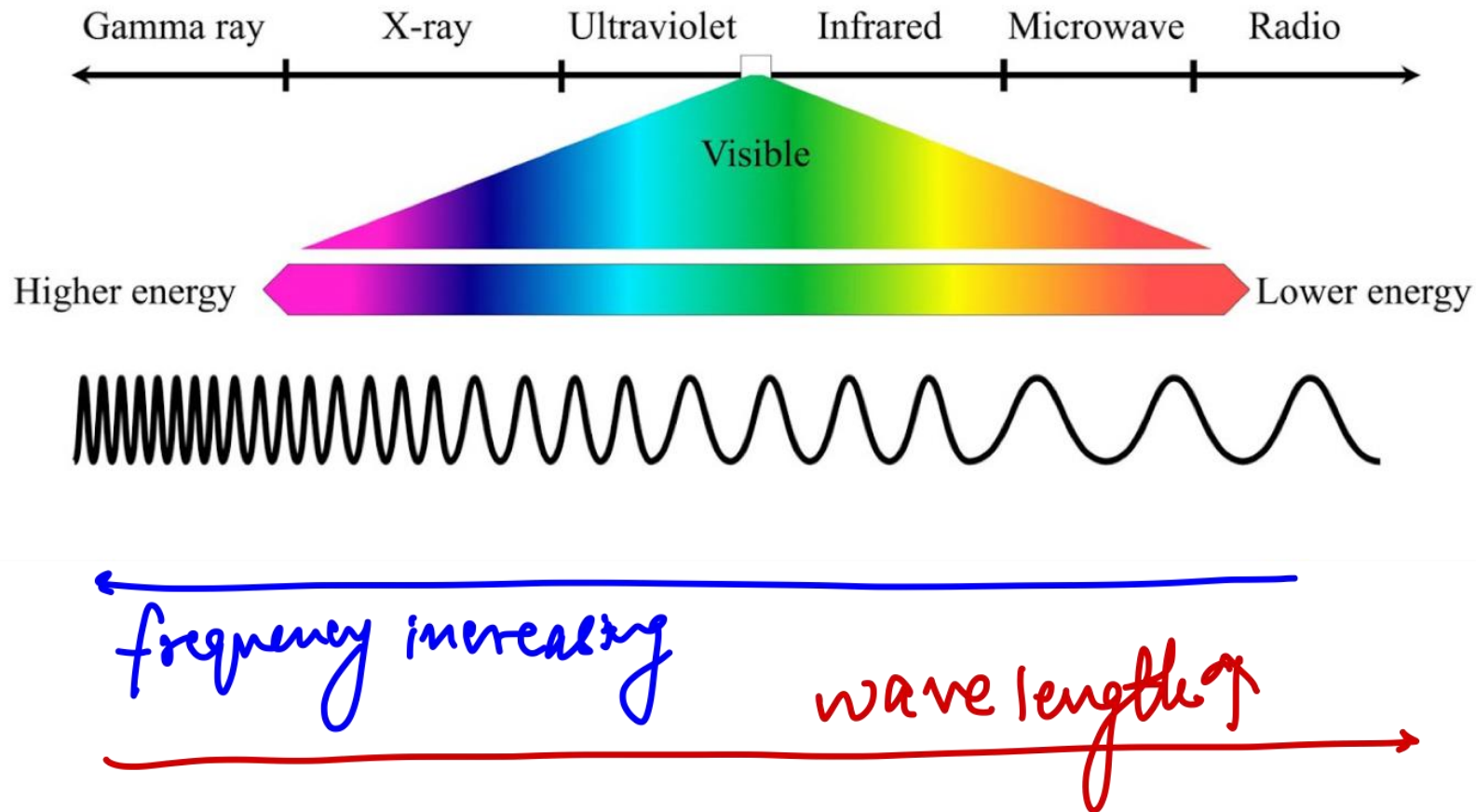
- a) Used for receiving and transmission on of signals.
- b) Many frequency bands : different forms of transmission due to different energy characteristics.
- c) Let's understand waves and the EM spectrum to understand this in detail.

Waves

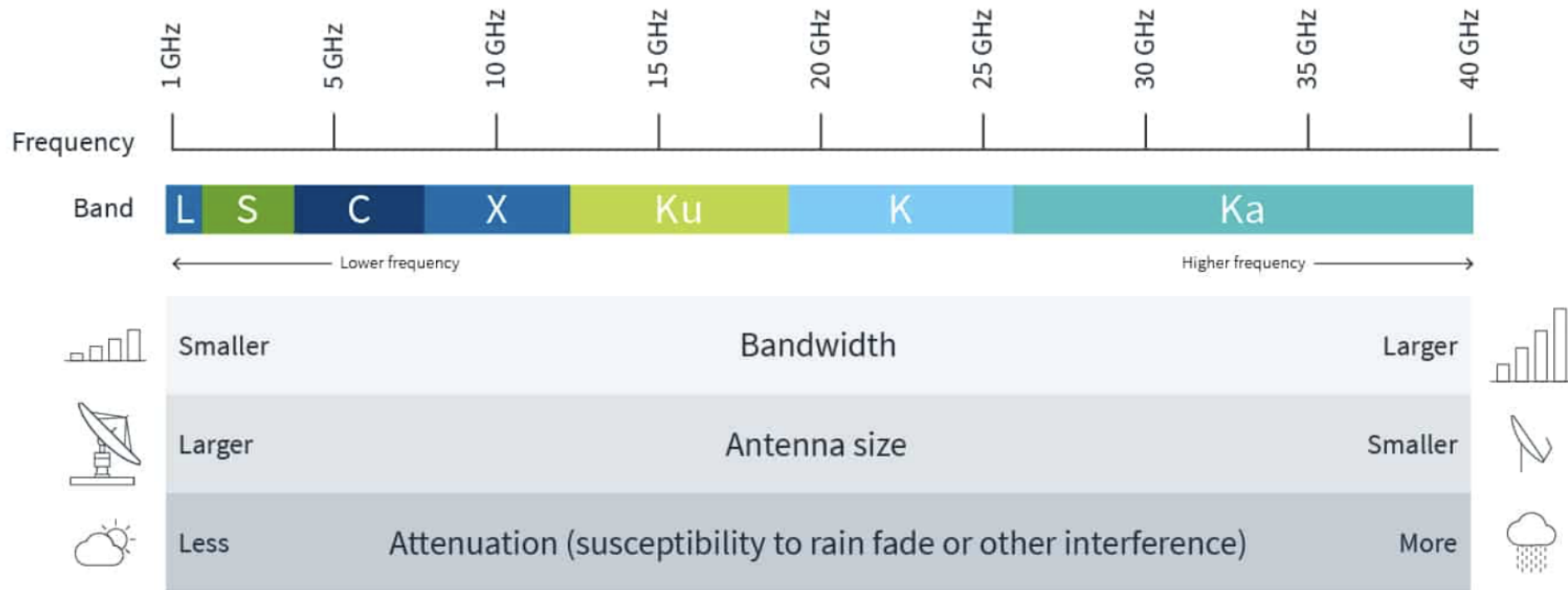
Some Basics :

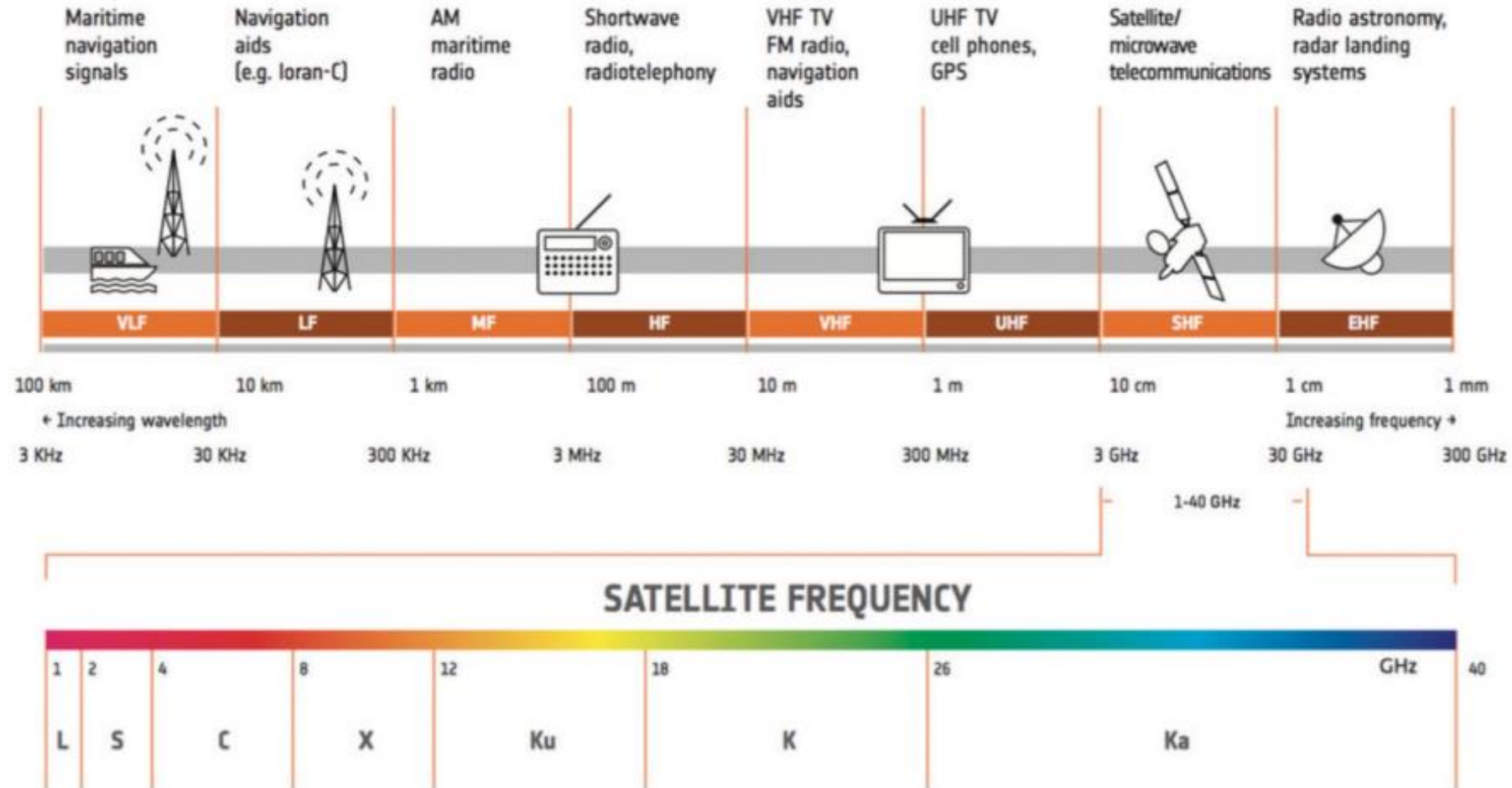
- Antenna Size is directly proportional to Wavelength.
- Frequency increase : more energy : more throughput (data transfer per unit time)
- Above 11Ghz : Wavelength are very small, Frequency is very high. Here there can be a phenomenon of RAIN FADE due to rain interfering with small wavelength particles.
- So we have to find the right balance.

Electromagnetic Spectrum



Satellite spectrum properties





Different Types of Bands

These are microwaves and radio waves

L-band (1–2 GHz)

Global Positioning System (GPS) carriers and also satellite mobile phones.

S-band (2–4 GHz)

Weather radar, surface ship radar, and some communications satellites, especially those of NASA for communication with ISS and Space Shuttle.

C-band (4–8 GHz)

Primarily used for satellite communications, for full-time satellite TV networks or raw satellite feeds. Commonly used in areas that are subject to tropical rainfall, since it is less susceptible to rain fade than Ku band

Different Types of Bands

X-band (8–12 GHz)

Primarily used by the military. X-band radar frequency sub-bands are used in civil, military and government institutions for weather monitoring, air traffic control, maritime vessel traffic control, defense tracking, and vehicle speed detection for law enforcement.

Ku-band (12–18 GHz)

Used for satellite communications

Ka-band (26–40 GHz)

Communications satellites for close up high resolutions applications, uplink in either the 27.5 GHz and 31 GHz bands, close-range targeting radars on military aircraft.

ISRO

Indian Remote Sensing Satellite (1988)

- (i) Survey and management of Natural resources. (Forest, agro, wetlands)
- (ii) Mineral prospecting : Locating minerals via Satellite (Resource SAT - 1, 2A, 2B)
- (iii) Cartography (Contour maps) : (Cartosat) (gis: geographical into system)

Bhuvan System : Collaboration between map my india and ISRO : Indigenous GeoSpatial Portal

IRS

(iv) Oceanography : mapping and study of geography.

Eg: OCEANSAT (4-5)

Scatsat-1 has replaced OCEANSAT 2-A

SARAL: Satellite for ARGUS and ALTIGA (ISRO) and CNES (france)

(v) Environment : Pollution management.

3 Countries have dedicate climate change satellites - US, China, Japan.

IRS

NISAR : observation of changes our earth (Land S band.)

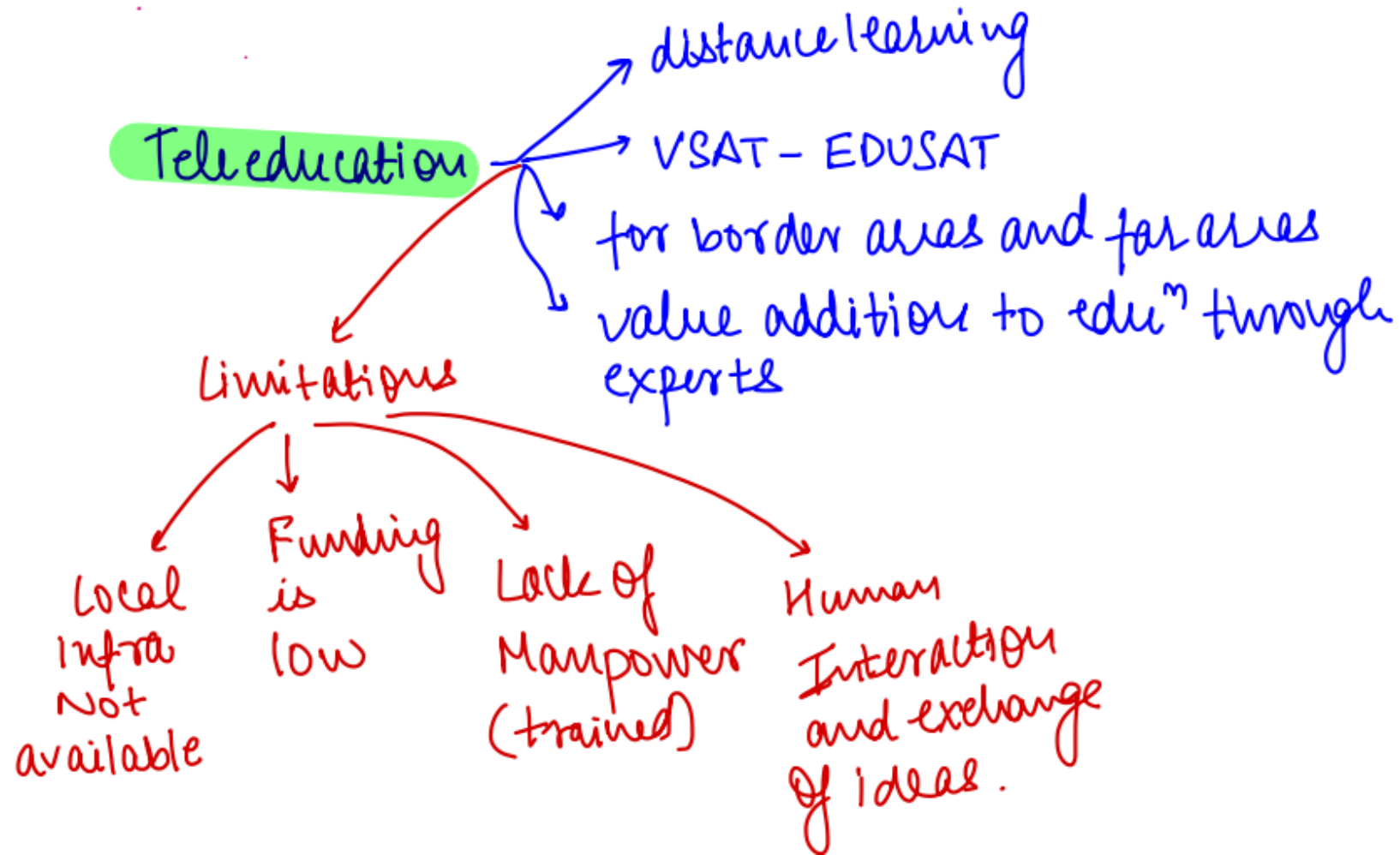
NISAR : Nasa ISRO Synthetic Aperture Radar : Collaboration between India and NASA for a dual frequency satellite.

(vi) Disaster Management : Hazard Mapping of disaster prone areas

INSAT : GSAT (1983)

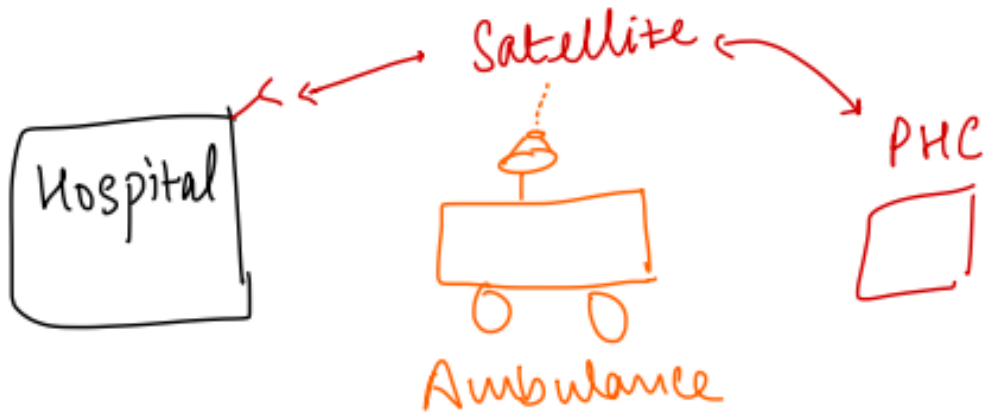
These satellites all in geostationary/geosynchronous orbit

- (i) Communication (all types)
- (ii) Weather forecasting (INSAT-3D) (INSAT-3DR)
- (iii) Tele education (Distance learning) (EDU-SAT)
- (iv) Telemedicine
- (v) Disaster Management



Telemedicine

: Healthcare services to rural via satellite communication



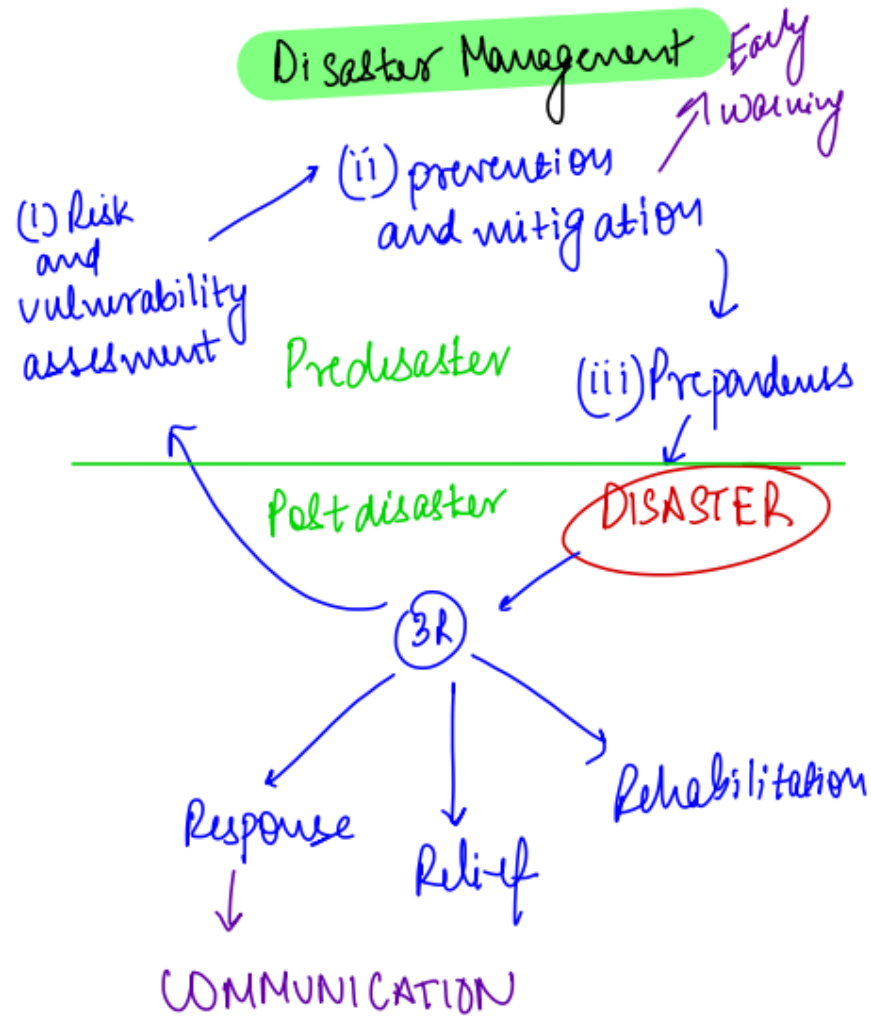
- 1st time done in Bangalore.
- already established system by ISRO.

Applications of Telemedicine

- Monitoring and treatment of conditions
- Continuing Medical education (CME).
- Emergency Situations: Trauma/RTA. Doctor can guide paramedics, (golden hour concept)
- Disaster Management.
- Aid to foreign countries - Outreach Programs
- **What is the Name of India's Flagship Telemedicine Program?**

Challenges

- No examination
- Psychology of Patients
- Digital Divide : Infrastructure issues
- Technical issues



- Mapping of hazard prone areas.
- Forecast and Warning
- Post Disaster Coordination
- DISANET : Network for Disaster management by Government of India.

GAGAN

- GPS aided geo augmented Navigation satellite system .
- AAI uses GAGAN for civil aviation and air traffic management safety management.

Use of 3 GSAT

Main Purpose is to prevent air line accidents and manage traffic.

Ex : MH 371 : Malaysian Airplane shot down.

Related News

- Airports Authority of India (AAI) successfully conducted flight trials using GAGAN based LPV approach procedure.
- India is the first country in the Asia Pacific Region to achieve such a landmark in field of Air Navigation Services (ANS).
- LPV (Localizer Performance with Vertical Guidance) permits aircraft guided approaches that are operationally nearly equivalent to Category 1- Instrument Landing System (Cat-1 ILS), without the need for ground-based navigational infrastructure.

About GAGAN (GPS Aided GEO Augmented Navigation)

- GAGAN is an Indian Satellite Based Augmentation System (SBAS) jointly developed by AAI and ISRO for India and neighbouring countries in the equatorial region.
- It is one among the only four Space-Based augmentation systems available in the world which also includes US(WAAS) Europe (EGNOS) and Japan (MSAS).
- Benefits of GAGAN: Air traffic control, manage road and railways transport, help farmers in crop spraying etc.

GPS aided geo augmented navigation

- ISRO+AAI
- Allows planes to land without ground markers :
 - Performance similar to cat-1 ILS
 - Instrument landing system
- Localizer performance with vertical guidance
- Only 4 countries : USA, Europe, Japan, India
- RUSSIA AND CHINA DONOT HAVE SUCH A SYSTEM

SPACE EXPLORATION

1. Orbit : Spacecraft orbits celestial body
2. Lander : Module used to land rover on a celestial body. (Soft landing) use of parachute.
3. Rover - Robotic device carries instruments as payload for scientific investigation of celestial bodies.

eg: Phoenix, Curiosity, Insight (for Mars)
by NASA
India - Chandrayan-2
Lander (VIKRAM)
Rover (PRAGYAN)

SPACE EXPLORATION

Fly-By Mission - Just pass by planet and collect data.

Eg: Voyager - I to jovian planets (1974). It was sent to study the outer edges of solar systems

Gravity Assist : Use of sun's gravity/other planets to propel space craft

ISRO Missions : Space Exploration

Earth's Moon : Chandrayaan 1 (2008), Chandrayaan 2 (2019) and Chandrayaan 3

Mars : Mangalyaan 1 and Mangalyaan 2

Astronomical Observation : AstroSat

Study of Sun : Aditya L1

Chandrayan - 1

- Orbiter only
- Had to be stopped due to battery issues
- Found loss of water are on surface of moon, confirmed by NASA
- Found landing site of apollo mission .
- Carried instruments from Multiple Countries.

Chandrayaan - 2 (3 parts)

- Orbiter
- Lander : Vikram
- Rover : Pragyan
- Assistance from Russia and France.
- Orbiter was successful, but lander and rover could not reach the surface of the moon.

Study of Moon

- Terrain and surface mapping
- Spectrometry of Soil of moon for Mineral Content
- Atmosphere of moon to check for any traces of gas.
- Presence of water on moon.

GSLV-Mk III-M1/ Chandrayaan-2 Mission

- India's Geosynchronous Satellite Launch Vehicle, GSLV Mk III-M1, successfully launched Chandrayaan-2 spacecraft on July 22, 2019 into its planned orbit with a perigee (nearest point to Earth) of 169.7 km and an apogee (farthest point to Earth) of 45,475 km. The launch took place from the Second Launch Pad at Satish Dhawan Space Centre SHAR Sriharikota.

GSLV-Mk III-M1/ Chandrayaan-2 Mission

- Chandrayaan-2 mission is a highly complex mission, which represents a significant technological leap compared to the previous missions of ISRO. It comprised an Orbiter, Lander and Rover to explore the unexplored South Pole of the Moon. The mission is designed to expand the lunar scientific knowledge through detailed study of topography, seismography, mineral identification and distribution, surface chemical composition, thermo-physical characteristics of top soil and composition of the tenuous lunar atmosphere, leading to a new understanding of the origin and evolution of the Moon.

GSLV-Mk III-M1/ Chandrayaan-2 Mission

- After the injection of Chandrayaan-2, a series of maneuvers were carried out to raise its orbit and on August 14, 2019, following Trans Lunar Insertion (TLI) maneuver, the spacecraft escaped from orbiting the earth and followed a path that took it to the vicinity of the Moon. On August 20, 2019, Chandrayaan-2 was successfully inserted into lunar orbit. While orbiting the moon in a 100 km hinar polar orbit, on September 02, 2019, Vikram Lander was separated from the Orbiter in preparation for landing. Subsequently, two de-orbit maneuvers were performed on Vikram Lander so as to change its orbit and begin circling the moon in a 100 km x 35 km orbit. Vikram Lander descent was as planned and normal performance was observed upto an altitude of 2.1 km. Subsequently communication from lander to the ground stations was lost.

GSLV-Mk III-M1/ Chandrayaan-2 Mission

- The Orbiter placed in its intended orbit around the Moon will enrich our understanding of the moon's evolution and mapping of the minerals and water molecules in Polar regions, using its eight state-of-the-art scientific instruments. The Orbiter camera is the highest resolution camera (0.3 m) in any lunar mission so far and will provide high resolution images which will be immensely useful to the global scientific community. The precise launch and mission management has ensured a long life of almost seven years instead of the planned one year.

Chandrayaan 3

- The space agency is aiming to achieve a soft landing on the south pole of the lunar surface, which is the least explored region of the Moon to date.
- Unlike its predecessor, Chandrayaan 3 will not carry an orbiter—but will include a lander and a rover to study the lunar surface

Countries with LUNAR missions

- (1) NASA= Apollo
- (2) Russia = LUNA Nussion
- (3) ESA
- (4) China : Change-E mission
- (5) Japan

Mangalyaan (MOM) - 1

- India's Mars Orbiter Mission (MOM) spacecraft has lost communication with the ground stations, bringing an end to its life after eight long years.

Mars Orbiter Mission

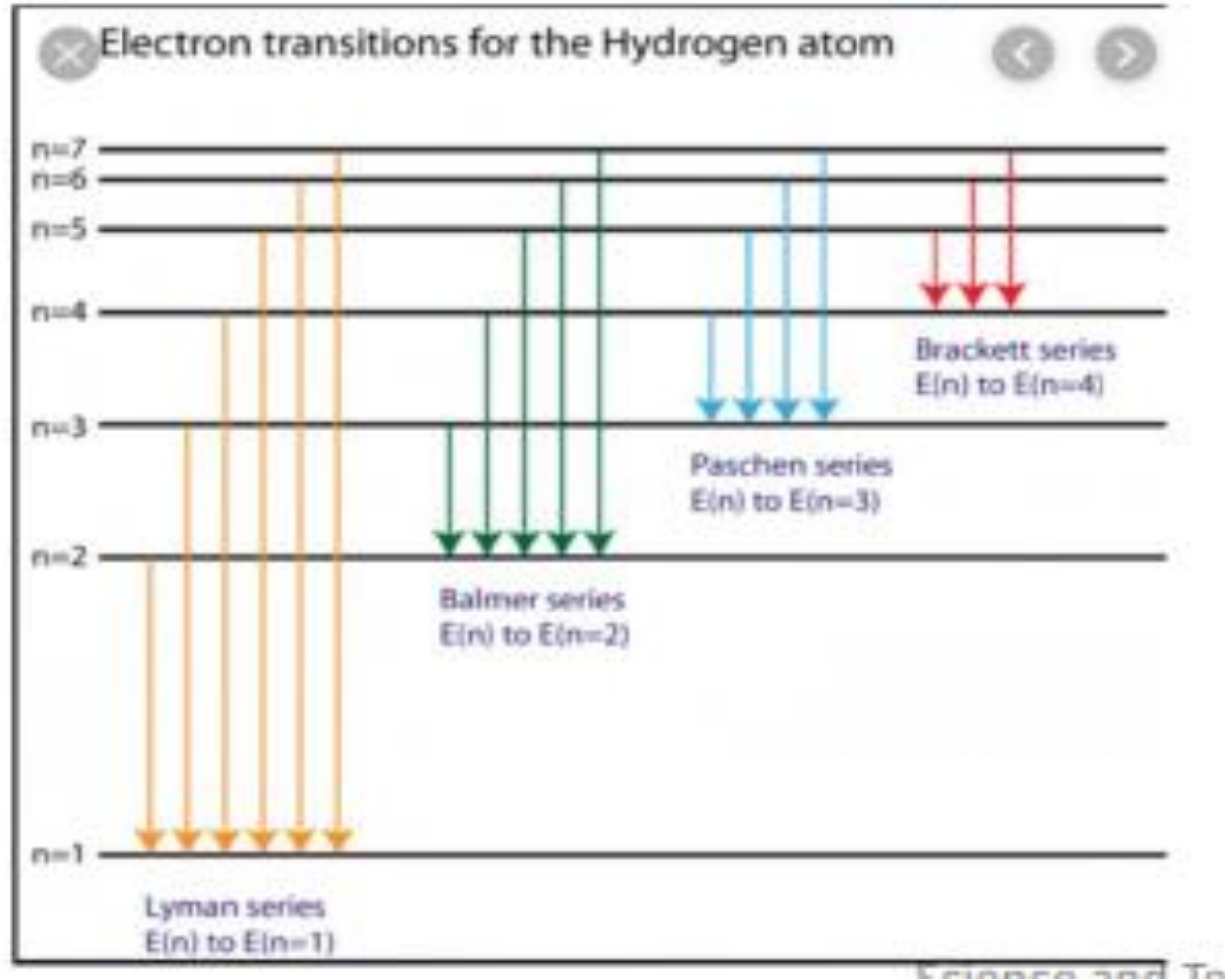
- (1) India's first interplanetary exploration mission/
- (2) Objectives
 - (a) Technology demonstration : interplanetary operation
 - (b) Scientific objective : data collection and exploration of martian surface features

Key fact about the mission

- Launched onboard PSLV-C25 (2013), It is India's first interplanetary mission
- Mission made India the first Asian country, and the fourth in the world (after Russia, US and EU) to get to the planet.
- Carried 850 kg of fuel and 5 science research payloads.
- Spacecraft is tracked by the Indian Deep Space Network (IDSN) located near Bengaluru

For scientific objective : 5 instruments

- (1) **Mars Color Camera** : Will take pictures of Mars' Surface and this will put other information into context.
- (2) **Lyman Alpha Photometer** : Ratio of Deuterium and Hydrogen. This will answer questions about water presence on Mars.
- (3) **Thermal Infrared imaging Spectrometer** : It will map the surface composition and Mineralogy of the planet by measuring thermal emission.
- (4) **MENCA** : mars exospheric neutral composition analyser. Seasonal Variations in the martian atmosphere.
- (5) **Methane Sensor** : Scan the entire martian disc in 6 minutes and measure very low levels of methane in parts per billion.

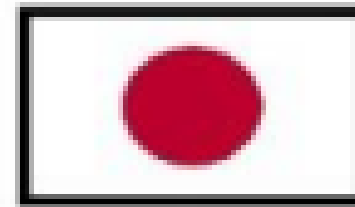


Technological Achievements of MOM

- Understanding of the composition of several gases in the martian exosphere including discovery of ‘suprathermal’ Argon-40 atoms.
- It Photographed, for the first time, the far side of Deimos, one of the natural satellites of Mars.
- Mission captured the time-variation of the Martian polar ice caps.

Other Mars Mission

- Perseverance (NASA, USA)
- Hope Mission (United Arab Emirates)
- Tianwen 1 (China)
- Mars Moons Exploration (MMX) Mission (Japan)

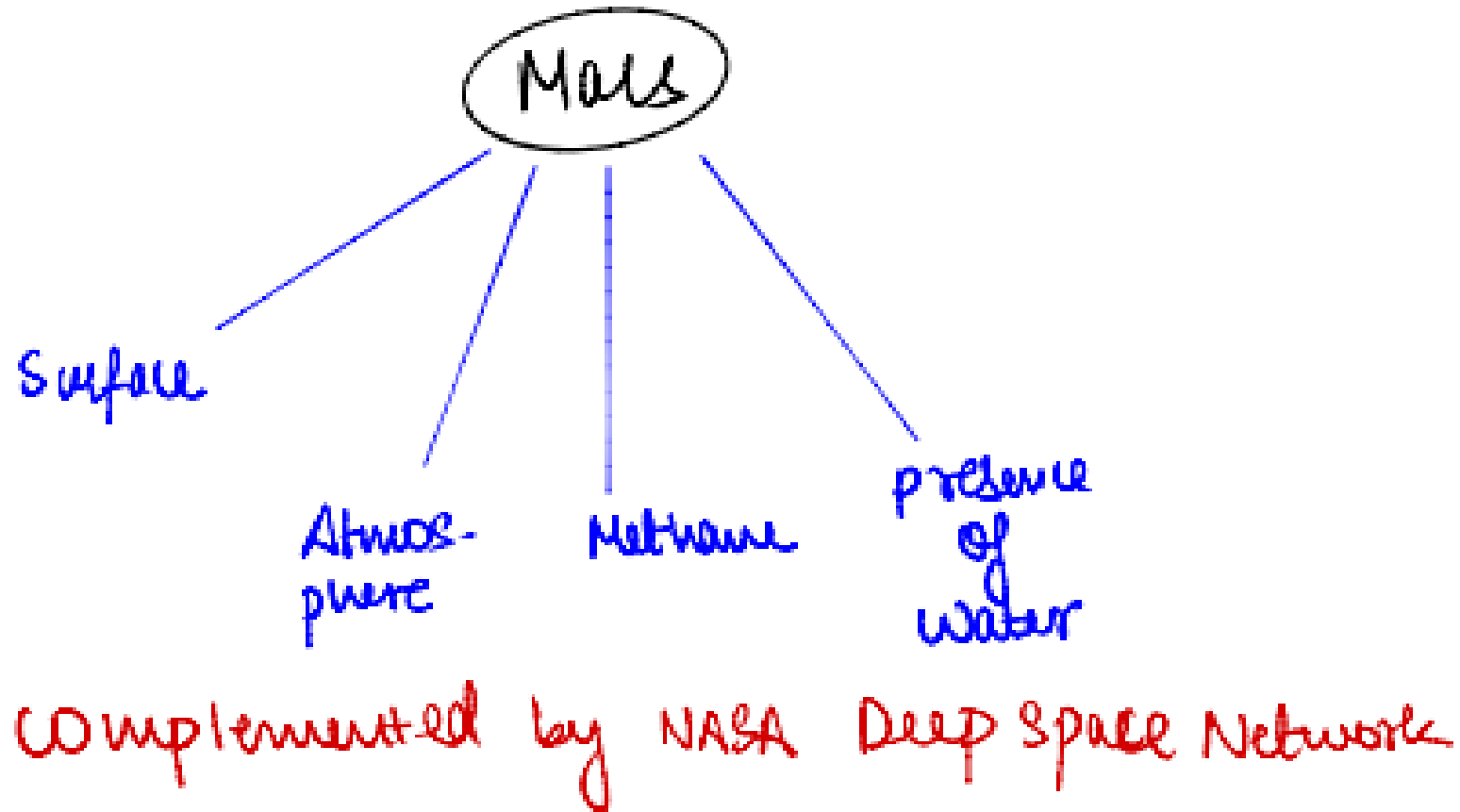


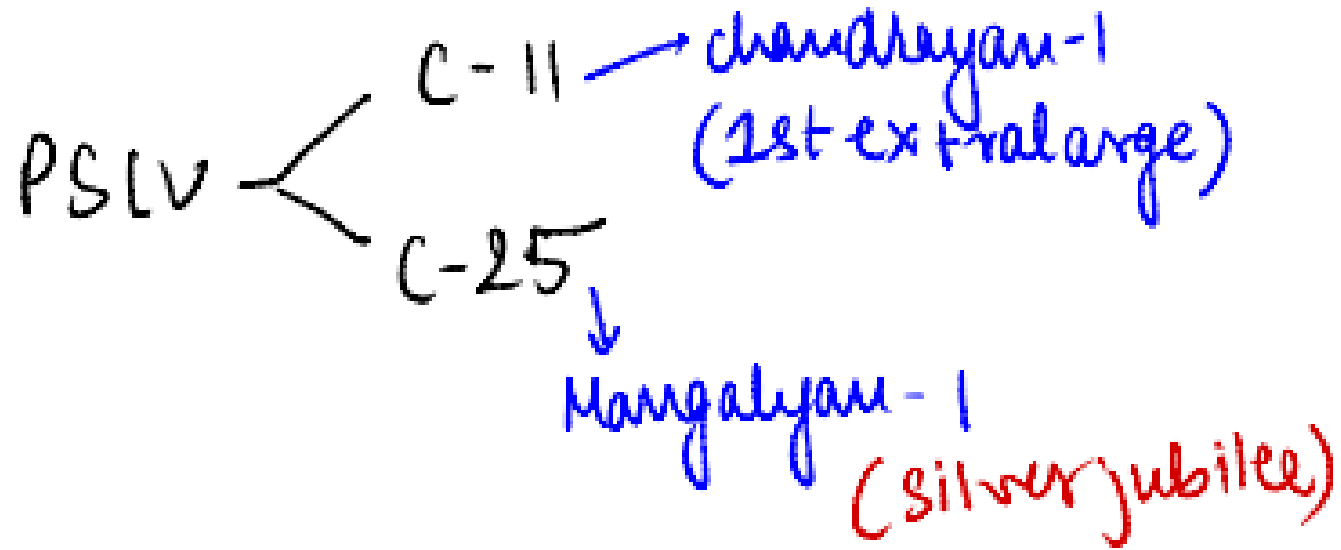
NASA's Perseverance Rover

- Perseverance rover mission is part of NASA's Mars Exploration Program, a long-term effort of robotic exploration of Mars.
- Recently, it placed a titanium tube containing a rock sample on the surface of Mars.
- Igneous rock sample was collected from Mars Jezero Crater called South Séítah.
- Samples are being placed at a location called "Three Forks," first such sample depot on another world.
- Depot will serve as a backup if Perseverance can't deliver its samples and subsequent NASA missions would

NASA's Perseverance Rover

- Recently, it also captured the solar eclipse on Mars featuring Phobos, one of Mars' two moons (other is Deimos).
- In a related news, Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE) has produced oxygen at Mars with components from the planet's atmosphere.
- MOXIE was sent (by Massachusetts Institute of Technology) with NASA's Perseverance rover





Mangalyan-2 : GSLV

ASTROSAT 2015

India's astronomical Satellite

① Multiwavelength spacecraft.

- visible
- x-ray
- UV rays

Orbit = 650 Km (10 W earth Orbit)

Scientific Objectives:

1. High energy events of universe
 - Supernova
 - Collapsing black hole
2. Binary system study (2 star system) quasars and pulsars.
3. Mapping universe for black holes and study them
 - NASA has Chandra X ray observatory to map for black holes
Named after Indian scientist
 - Other countries with astronomical Satellite
 - Japan, ESA, Many More

Scientific Objectives:

- ASTROSAT is India's first dedicated multi wavelength space observatory. This scientific satellite mission endeavours for a more detailed understanding of our universe. One of the unique features of ASTROSAT mission is that enables the simultaneous multi-wavelength observations of various astronomical objects with a single satellite.

Scientific Objectives:

- ASTROSAT observes universe in the optical, Ultraviolet, low and high energy X-ray regions of the electromagnetic spectrum, whereas most other scientific satellites are capable of observing a narrow range of wavelength band. Multi-wavelength observations of ASTROSAT can be further extended with co-ordinated observations using other spacecraft and ground based observations. All major astronomy Institutions and some Universities in India are participating in these observations.

Scientific Objectives:

- ASTROSAT with a lift-off mass of about 1513 kg was launched into a 650 km orbit inclined at an angle of 6 deg to the equator by PSLV-C30. After injection into Orbit, the two solar panels of ASTROSAT were automatically deployed in quick succession. The spacecraft control centre at Mission Operations Complex (MOX) of ISRO Telemetry, Tracking and Command Network (ISTRAC) at Bangalore manages the satellite during its mission life.

Scientific Objectives:

- The science data gathered by five payloads of ASTROSAT are telemetered to the ground station at MCX. The data is then processed, archived and distributed by Indian Space Science Data Centre (SSDC) located at Byalalu, near Bangalore.

The scientific objectives of ASTROSAT mission are:

- To understand high energy processes in binary star systems containing neutron stars and black holes
- Estimate magnetic fields of neutron stars
- Study star birth regions and high energy processes in star systems lying beyond our galaxy
- Detect new briefly bright X-ray sources in the sky
- Perform a limited deep field survey of the Universe in the Ultraviolet region

Payloads of ASTROSAT:

- Five payloads of ASTROSAT are chosen to facilitate a deeper insight into the various astrophysical processes occurring in the various types of astronomical objects constituting our universe. These payloads rely on the visible, Ultraviolet and X-rays coming from distant celestial sources.

Payloads of ASTROSAT:

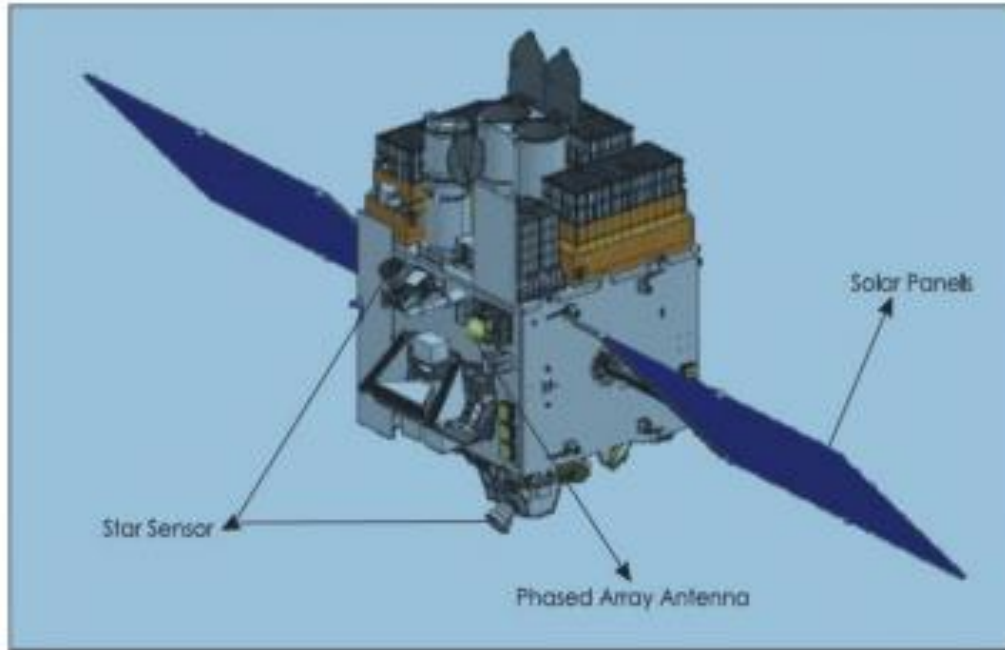
- 1. The Ultraviolet Imaging Telescope (UVM), capable of observing the sky in the Visible, Near Ultraviolet and Far Ultraviolet regions of the electromagnetic spectrum
- 2. Large Area X-ray Proportional Counter (LAXPC), is designed for study the variations in the emission of X-rays. from sources like X-ray binaries, Active Galactic Nuclei and other cosmic sources.

Payloads of ASTROSAT:

- 3. Soft X-ray Telescope (SXT) is designed for studying how the X-ray spectrum of 0.3-8 keV range coming from distant celestial bodies varies with time,
- 4. Cadmium Zinc Telluride Imager (CZTI), functioning in the X-ray region, extends the capability of the satellite to sense X-rays of high energy in 10-100 keV range.

Payloads of ASTROSAT:

- 5. Scanning Sky Monitor(SSM), is intended to scan the sky for long term monitoring of bright X-ray sources in binary stars, and for the detection and location of sources that become bright in X-rays for a short duration of time



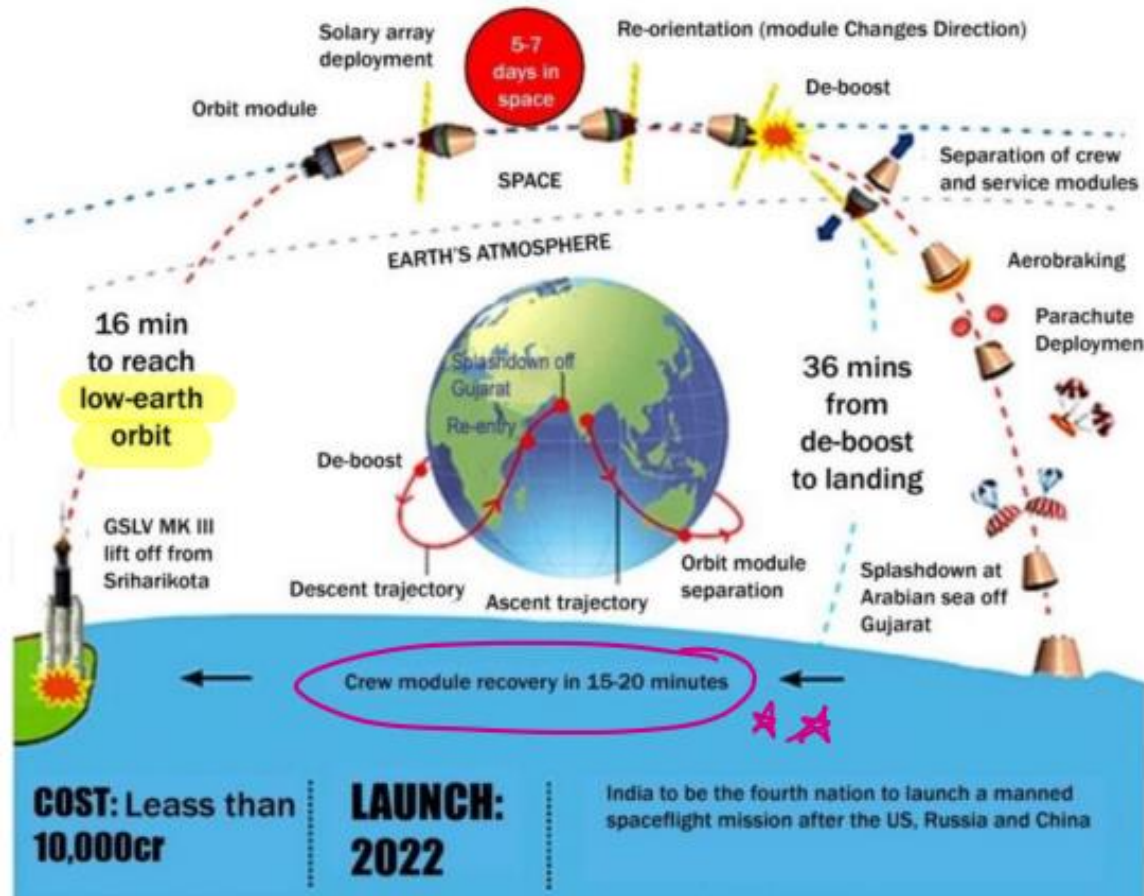
Gaganyaan

- The Gaganyaan orbital module has two parts the Crew Module (CM) and the Service Module (SM) and weighs about 8000 kg.
- The Crew Module (CM) is a double walled system and a habitat of astronauts. It has an ablative thermal protection system (TPS) to protect it during the intense aerodynamic heating during the flight.

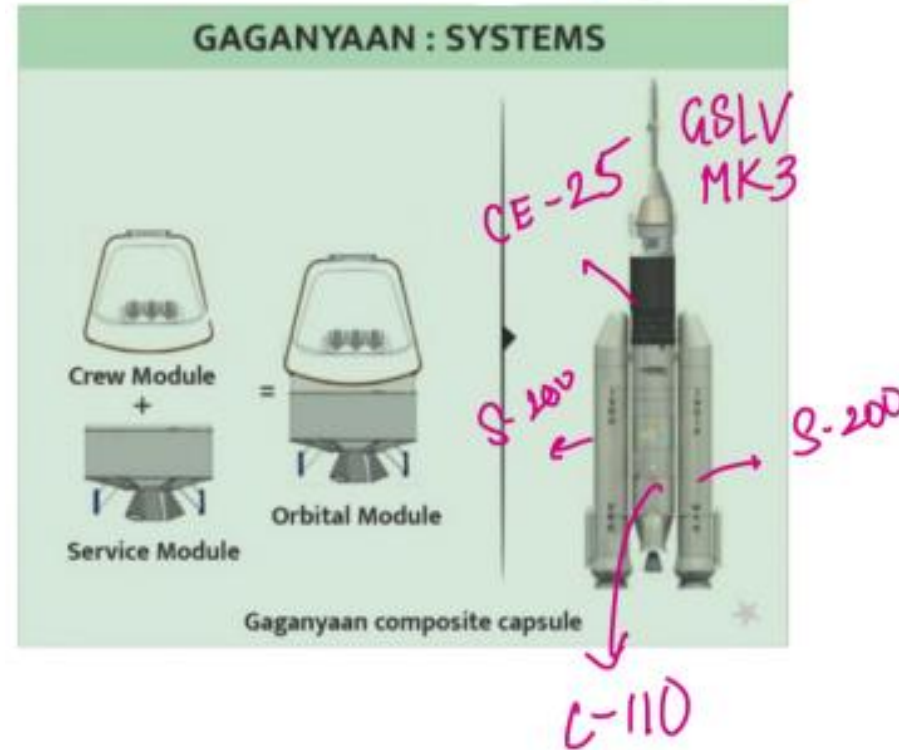
Gaganyaan

- The Orbital module (OM) will be launched by a human-rated GSLV MK-III vehicle and will orbit the Earth with a velocity of about 7,800 metre/second.
- Gaganyaan would be smaller in size than the current manned missions of Russian Soyuz, Chinese Shenzhou and NASA's planned Orion spacecraft.

MANNED MISSION

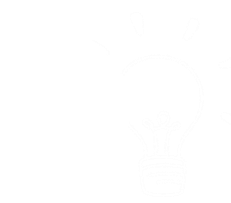


GAGANYAAN : SYSTEMS



SSLV

- Small Satellite Launch Vehicle (SSLV)
- It would help to meet the demand of the global launch services market for small satellites (nano, micro etc).
- New Space India Limited (NSIL), a PSU, will be the sole nodal agency responsible for providing end-to end SSLV Launch services.



Private Sector in Space

- Government has created Indian National Space Promotion and Authorization Centre (INSPACe) to boost private sector participation in entire range of space activities.
- About IN-SPACe It is the new entity of the Department of Space which will have its own chairperson and board. It will regulate and promote building of routine satellites, rockets and commercial launch services through Indian industry and startups. It will have its own directorates for technical, legal, safety and security, monitoring and activities promotion.

Private Sector in Space

- It will act as an interface between ISRO and private parties, and assess how best to utilise India's space resources and increase space-based activities.
- It will function autonomously and parallel to ISRO. ISRO will remain the basic body that decides what missions are to be undertaken but INSPACe will help fill the gaps

NSIL

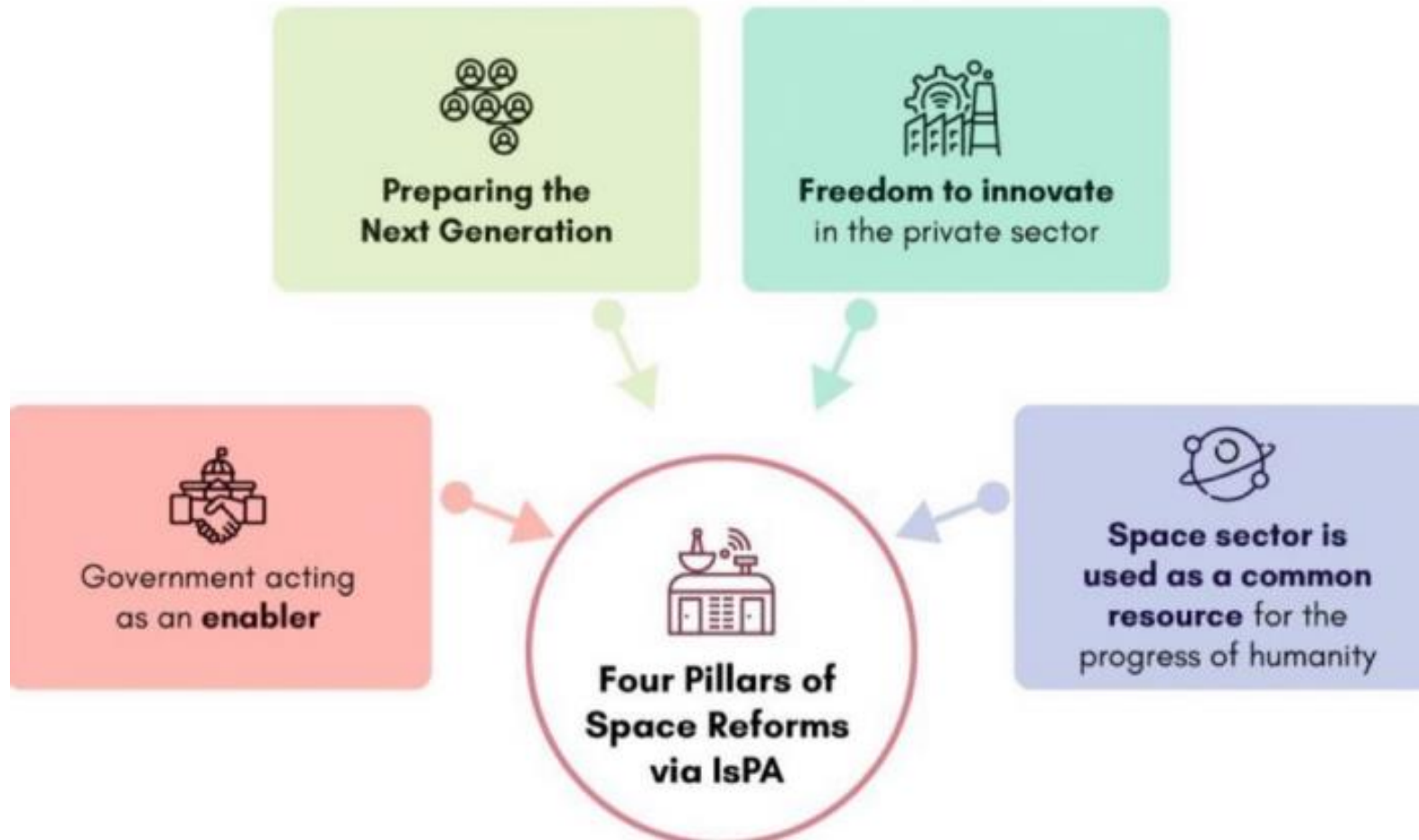
- About New Space India Limited
- It was founded in 2019, by Department of Space (DOS) to bridge gap between ISRO and private sector and facilitates transfer of ISRO technologies to industry.
- It is a wholly-owned Government of India company, under the administrative control of DOS.
- It will not replace ANTRIX, a private limited company owned by Government of India in 1992 as a Marketing arm of ISRO

Indian Space Association

- The industry association will act as an independent and a “single-window” agency for enabling the opening up of the space sector to start-ups and the private sector.
- ISpA will also work towards building global linkages for the Indian space industry to bring in critical technology and investments into the country to create more high-skill jobs
- The current global space economy is estimated at US\$ 360 billion, with India accounting for ~2% (US\$ 7 billion).

Indian Space Association

- In the global space market, rocket and satellite launch services, an area in which ISRO specialises, amount to only 5% share. Satellite-based services and ground-based systems account for the remaining 95%.
- Several private companies are developing plans to take paying customers to space on a regular basis leading to a rapid growth in space tourism industry.



Recent initiatives by Private Sector

- Hyderabad's Dhruva Space and Bengaluru's Digantara were authorized by IN-SPACe for launch of payloads onboard PSLV Orbital Experimental Module (POEM) of PSLV-C53.
- Mission Prarambh: Involved launch of Vikram-S (VKS), India's first privately built rocket by Hyderabadbased Skyroot Aerospace.
- VKS is solid propellant rocket.
- Payload Capacity: 290 kg- 560 kg support from ISRO and IN-SPACe

Recent initiatives by Private Sector

- India's first private launch pad unveiled at Sriharikota
- Agnikul and executed in support of ISRO and IN-SPACe.
- It is specifically built to support liquid-stage controlled launches.

Space Tourism

- Space tourism is space travel for recreational, leisure or business purposes. There are several different types of space tourism, including orbital, suborbital and lunar space tourism.
- To date, orbital space tourism has been performed only by the Russian Aviation and Space Agency- Roscosmos.
- Several private spaceflight companies are now working towards developing suborbital space tourism vehicles to take paying customers to space.

Space Tourism

- No international space law has defined space tourists. Existing space treaties such as Outer Space Treaty, Rescue Agreement etc. are only applicable to astronauts, envoys of mankind, or personnel of a spacecraft.
- To send a vehicle to space, it is compulsory for the national and international space law to authorize it according to Article VI of the Outer Space Treaty. However, there is lack of clarity on authorization of space travel with tourists aboard.

Space Travel

- According to the Fédération Aéronautique Internationale (world governing body for aeronautic and astronautic records), space starts at an altitude of 100 km (62 miles) above the surface of the earth.
- This is the Karman line where atmospheric lift no longer supports a flying object and the object would need to reach orbital velocity or risk falling back to Earth.

Space Travel

- Space travel is referred to as any flight operation that takes one or more passengers beyond the altitude of 100 km and thus into space.
- The main difference between orbital and suborbital flight is the speed at which a vehicle is traveling.
- An orbital spacecraft must achieve orbital velocity i.e. the speed that an object must maintain to remain in orbit around a planet. To orbit 125 miles (200 kilometers) above Earth for instance, a spacecraft must travel at a screaming 17,400 mph (28,000 km/h).

Space Travel

- Suborbital flight, in contrast, requires much lower speeds and doesn't have the power to achieve orbit. Instead, it will fly up to a certain height that depends on its speed, and then come back down once its engines are shut off.
- At the top of their flight arc in a suborbital flight, when the object is falling back toward Earth, passengers achieve a few minutes of weightlessness under zero gravity.

